Lecture Scribe CSE400

L15_L17.pdf
PDF

I want a lecture scribe for the pdf i gave you so that i can refer it on the exam day for revision
and also use only the content given in the slide.do not use anyother information from outside
the scribe should include :
1.)definations
2.)symbols
3.)important results
4.)formula
5.)symbols
6.)proof
7.)derivation should be step by step and same with examples.
the scribe should not include new explanations,example,and also do not simplify or modify anything from slide.Do not ask me any question give the respose dierctly

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CSE400 – Lecture L15–L18 Scribe

Joint Random Variables + Joint Distributions

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1. DEFINITIONS

Joint Random Variables

- Two (or more) random variables considered at the same time.

Discrete Case – Joint PMF

Setup

- Suppose X and Y are discrete random variables
- $X = \{x_1, x_2, \ldots, x_n\}$
- $Y = \{y_1, y_2, \ldots, y_m\}$
- Ordered pairs:

$$(X, Y) \in \{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

- Probability of joint outcome occurring simultaneously

Continuous Case – Joint PDF

Definition (Joint PDF)

A function $f(x, y)$ is called the joint probability density function if:

$$P((X, Y) \text{ in small rectangle}) \approx f(x, y)\, dx\, dy$$

Joint CDF

Notation

$$X \leq x,\ Y \leq y$$

Means:

$$\{X \leq x \text{ and } Y \leq y\}$$

Definition (Joint CDF)

$$F(x, y) = P(X \leq x, Y \leq y)$$

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2. SYMBOLS

- X, Y → Random variables
- x_i, y_j → Discrete values
- $p(x_i, y_j)$ → Joint PMF
- $f(x, y)$ → Joint PDF
- $F(x, y)$ → Joint CDF
- dx, dy → Infinitesimal intervals
- A → Event

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3. IMPORTANT RESULTS

Discrete Joint PMF Properties

$$0 \leq p(x_i, y_j) \leq 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

Continuous Joint PDF Properties

$$f(x, y) \geq 0$$

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

- Joint PDF may take values greater than 1
- It is a density, not a probability

Joint CDF (Continuous)

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Recovering PDF

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

Joint CDF (Discrete)

$$F(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

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4. FORMULAS

Joint PMF

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Joint PDF

$$P = f(x,y)\, dx\, dy$$

Normalization (Continuous)

$$\int_c^d \int_a^b f(x,y)\, dx\, dy = 1$$

Joint CDF

$$F(x,y) = P(X \le x, Y \le y)$$

Joint CDF (Continuous)

$$F(x,y) = \int_c^y \int_a^x f(u,v)\,du\,dv$$

Recover PDF

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

Joint CDF (Discrete)

$$F(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

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5. PROOF

Validity of Joint PDF (Example)

Given:

$$f(x,y) = 4xy, \quad 0 \le x \le 1,\ 0 \le y \le 1$$

Step 1: Normalization

$$\int_0^1 \int_0^1 4xy\,dx\,dy = \int_0^1 \left(\int_0^1 4xy\,dx \right) dy = \int_0^1 2y\,dy = [y^2]_0^1 = 1$$

Hence:

- $f(x,y) \ge 0$
- Total probability = 1
- Valid joint PDF

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6. DERIVATION (STEP-BY-STEP)

Compute $P(A)$

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

Step 1

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

Step 2

$$= \int_0^{0.5} \left[2xy^2\right]_{0.5}^{1} dx$$

Step 3

$$= \int_0^{0.5} 2x(1 - 0.25) \, dx$$

Step 4

$$= \int_0^{0.5} 2x(0.75) \, dx = \int_0^{0.5} \frac{3}{2} x \, dx$$

Step 5

$$= \left[\frac{3}{4}x^2\right]_0^{0.5} = \frac{3}{16}$$

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7. EXAMPLE (FROM SLIDES)

Rolling Two Dice

- Experiment: Roll two fair dice

- X = value on first die
- T = total (sum)

Each outcome:

$$P(i,j) = \frac{1}{36}$$

Joint Table Observations

- Entries are 0 or $1/36$
- X and T are **not independent**

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8. ADDITIONAL RESULTS (FROM SLIDES)

Continuous Case Mapping

- Discrete values → Continuous intervals
- Joint PMF → Joint PDF
- Sums → Double integrals

Region

If:

$$X \in [a,b],\ Y \in [c,d]$$

Then:

$$(X,Y) \in [a,b] \times [c,d]$$

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9. PROPERTIES OF JOINT CDF

1. **Non-decreasing**

$$F(x,y) \text{ increases or stays constant}$$

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2. **Lower-left corner**

$$F(x, y) = 0 \quad \text{as } (x, y) \to (-\infty, -\infty)$$

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3. **Upper-right corner**

$$F(x, y) = 1 \quad \text{as } (x, y) \to (\infty, \infty)$$

END OF SCRIBE

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